

UNIVERSITY OF ARIZONA
DEPARTMENT OF EPIDEMIOLOGY AND BIOSTATISTICS
BIOS 648: HIGH-DIMENSIONAL HEALTH DATA ANALYSIS AND MACHINE
LEARNING

From Nuclei to Patients: A Systematic Evaluation of Aggregation Strategies for Bladder Cancer Recurrence Prediction Using Karyometric Morphometry Data

Enock Adu Bonsu

PhD Student, Biostatistics

enocka@arizona.edu

*Submitted in partial fulfillment of the requirements
for BIOS 648 — Spring 2026*

May 20, 2026

Keywords: bladder cancer recurrence, karyometry, multiple-instance learning, feature aggregation, cross-validation, ensemble methods, nuclear morphometry, Leakage

Abstract

Karyometric data from bladder cancer patients present a standard multiple-instance learning (MIL) problem: 92 morphometric features are extracted from approximately 100 cell nuclei per patient, yet the clinical outcome (recurrence vs. non-recurrence) is observed only once per patient. Standard practice collapses nuclear measurements to per-feature means, discarding distributional information, and frequently evaluates models through nucleus-level cross-validation, a method that introduces severe information leakage. We propose and systematically evaluate a three-layer framework comprising three patient-level representation strategies (mean + quantile summaries, score-guided softmax-weighted features, and probability aggregation) crossed with four classifiers (lasso logistic regression, random forest, PCA + LDA, and a single hidden-layer neural network), plus a stacking meta-ensemble; 13 models in total, all evaluated under rigorous patient-level leave-one-out cross-validation ($n = 84$ patients).

The score-guided softmax-weighted representation (Representation B) consistently outperformed mean-summary representations (Representation A) by 10–13 percentage points in misclassification error (MCE) across all four classifiers, demonstrating that the patient-level aggregation strategy contributes more to predictive performance than the choice of downstream classifier. The stacking meta-ensemble achieved the best overall performance (MCE = 0.321, 95% CI: 0.224–0.432; sensitivity = 0.641; specificity = 0.711), correctly identifying 25 of 39 recurrent patients. We further demonstrate, through a deliberate controlled experiment, that nucleus-level cross-validation inflates apparent performance by 33.9 percentage points (56 % overoptimism), confirming this as a critical methodological pitfall in hierarchical biomedical data. Feature importance analysis identified within-patient variability measures, specifically the standard deviation and 90th percentile of features f54 and f53, as the most predictive karyometric attributes, suggesting that nuclear morphological heterogeneity drives recurrence prediction more than mean morphology alone.

Table of Contents

1	Introduction	1
2	Methods	1
2.1	Data and Pre-processing	1
2.2	Patient-Level Representations (Layer 1)	2
2.2.1	Representation A: Mean + Quantile Summary Vector	2
2.2.2	Representation B: Score-Guided Instance-Weighted Vector	2
2.2.3	Representation C: Nucleus-Level Probability Aggregation	3
2.3	Patient-Level Classifiers (Layer 2)	3
2.4	Stacking Meta-Ensemble (Layer 3)	4
2.5	Evaluation Protocol	5
2.5.1	Patient-Level Leave-One-Out Cross-Validation	5
2.5.2	Performance Metrics and Uncertainty Quantification	5
2.5.3	Statistical Comparison and Leakage Demonstration	5
3	Results	6
3.1	Overview	6
3.2	Primary Outcome: Misclassification Error	7
3.3	Effect of Patient Representation	7
3.4	Effect of Classifier Choice	8
3.5	Sensitivity, Specificity, and Confusion Matrices	8
3.6	Additional Diagnostics	10
4	Discussion and Conclusions	10
4.1	Principal Findings	10
4.2	Interpretation of Feature Importance	11
4.3	Limitations	11
	References	12
S1	Supplementary Methods	13
S1.1	Neural Network Weight Budget	13
S1.2	Handling Near-Zero-Variance Features in LDA	13
S2	Supplementary Figures	14

1. Introduction

Bladder cancer recurs in over 50% of patients within five years of surgical resection [1], making accurate recurrence prediction clinically important for triaging surveillance and adjuvant therapy. Karyometry, computational measurement of nuclear morphology from histological images has been proposed as a non-invasive prognostic biomarker [1], but the optimal statistical approach for patient-level prediction from nucleus-level measurements remains open.

The core challenge is a measurement–outcome mismatch that is; morphometric features are extracted nucleus-by-nucleus (≈ 100 per patient), yet the clinical label, recurrent (R) or non-recurrent (NR), is observed once per patient, the defining structure of a *multiple-instance learning* (MIL) problem [3]. Standard practice collapses measurements into per-feature means [1], discarding distributional shape that may be more discriminating if recurrence is driven by a subpopulation of aberrant cells [10]. A second widespread flaw is performing cross-validation at the nucleus level rather than the patient level [9], which introduces severe leakage and inflates apparent performance.

We address both issues through a three-layer framework, three patient-level representation strategies (including a proposed score-guided softmax-weighted aggregation inspired by attention-based MIL [5]) crossed with four classifiers and combined via a stacking meta-ensemble, evaluated under rigorous patient-level leave-one-out cross-validation with an empirical quantification of leakage bias. The primary question is: *does the nucleus-to-patient aggregation strategy matter as much as the choice of downstream classifier?*

2. Methods

2.1 Data and Pre-processing

The karyometry dataset [1] comprises 84 bladder cancer patients: 39 with one or more recurrences (R) and 45 without recurrence during at least eight years of follow-up. Approximately 100 nuclei from the original tumor were imaged for each patient, yielding 8,427 nucleus-level observations in total. For each nucleus, 92 morphometric features (f1–f92) were extracted, including total optical density (OD), nuclear area, nuclear roundness, OD standard deviation, min/max axis ratio, OD histogram features,

co-occurrence matrix statistics, and run-length matrix statistics.

All 92 features were standardized to zero mean and unit variance using statistics computed from training-fold data only within each cross-validation fold, preventing any information from the held-out test patient from influencing feature scaling. No missing values were present. The mild class imbalance (39 R, 45 NR; 46.4% vs. 53.6%) was managed by reporting sensitivity and specificity separately in addition to overall MCE.

2.2 Patient-Level Representations (Layer 1)

Let x_{ijk} denote the value of feature k for nucleus j of patient i , where $j = 1, \dots, n_i$ ($n_i \approx 100$). We construct three patient-level representations of increasing sophistication.

2.2.1 Representation A: Mean + Quantile Summary Vector

For each feature k , we compute five summary statistics across the n_i nuclei, the mean, standard deviation, and the 10th, 50th (median), and 90th percentiles, yielding a $92 \times 5 = 460$ -dimensional patient vector:

$$\mathbf{a}_{ik} = \left(\bar{x}_{i.k}, s_{i.k}, Q_{10}(x_{i.k}), Q_{50}(x_{i.k}), Q_{90}(x_{i.k}) \right). \quad (1)$$

This is the standard approach in the karyometry literature, capturing distributional shape via quantiles and variability via the standard deviation. It corresponds to *uniform pooling* in which all nuclei contribute equally to the patient summary.

2.2.2 Representation B: Score-Guided Instance-Weighted Vector

The second representation replaces uniform weights with nucleus-specific weights derived from a recurrence-risk scorer, motivated by the possibility that a small subset of aberrant nuclei may drive recurrence risk. A Random Forest scorer is trained at the nucleus level using the patient’s group label as a pseudo-label for each nucleus (the standard MIL assumption). The scorer is re-fitted within each LOOCV fold and never sees the held-out patient’s nuclei, so nucleus-level overfitting does not leak across patients. It produces predicted recurrence probabilities $\hat{p}_{ij}$ for each nucleus.

Softmax-normalized instance weights with temperature $\tau > 0$:

$$w_{ij} = \frac{\exp(\hat{p}_{ij}/\tau)}{\sum_{j'} \exp(\hat{p}_{ij'}/\tau)}, \quad (2)$$

produce the patient representation:

$$b_{ik} = \sum_j w_{ij} x_{ijk}. \quad (3)$$

As $\tau \rightarrow \infty$, weights become uniform (recovering Representation A); as $\tau \rightarrow 0$, the highest-risk nucleus dominates, approximating max-pooling. The optimal $\tau \in \{0.1, 0.5, 1, 2, 5, \infty\}$ is selected by internal 5-fold CV within each LOOCV training fold. This mechanism is conceptually equivalent to attention-based pooling in deep MIL [5], adapted for the tabular small-sample setting where end-to-end attention training is statistically infeasible.

2.2.3 Representation C: Nucleus-Level Probability Aggregation

The third representation collapses the nucleus-level predicted probabilities to three scalar summaries:

$$\mathbf{c}_i = \left(\bar{\hat{p}}_{i\cdot}, \max_j \hat{p}_{ij}, Q_{90}(\hat{p}_{i\cdot}) \right), \quad (4)$$

yielding a 3-dimensional patient representation. This is intentionally a low-dimensional diagnostic extreme: if its performance matches Representation A (460 features), the richer distributional summaries carry little additional information.

2.3 Patient-Level Classifiers (Layer 2)

Four classifiers are applied to each representation.

Lasso logistic regression. For patient i with representation $\mathbf{z}_i$ and outcome $y_i \in \{0, 1\}$:

$$\log \frac{P(y_i = 1 | \mathbf{z}_i)}{P(y_i = 0 | \mathbf{z}_i)} = \beta_0 + \boldsymbol{\beta}^\top \mathbf{z}_i, \quad (5)$$

with L_1 penalty $\lambda \sum_k |\beta_k|$. The penalty λ is selected by 5-fold CV via `glmnet` [4]; automatic variable selection is critical for Representation A (460 features, 83 training patients).

Random Forest. 500 trees with $\lfloor \sqrt{p} \rfloor$ split features via `randomForest` [6]. Permutation importance is extracted for interpretability.

PCA + LDA. For Representation A, PCA reduces the 460-dimensional space to M principal components (chosen by 5-fold CV, $M \leq 30$), then LDA is applied. For Representations B and C, LDA is applied directly. Features with within-class SD $< 10^{-8}$ are removed to prevent singularity.

Neural network. Single hidden layer, H hidden units ($H \in \{2, 4, 6, 8, 10\}$, capped by a 3,000-weight budget) and weight decay $\lambda \in \{0.001, 0.01, 0.1\}$ via `nnet` [7]. Five random restarts per (H, λ) ; best-fitting solution retained.

The resulting $3 \times 4 = 12$ models (M1–M12) plus the majority-class baseline (M0) are summarized in Table 1.

Table 1. Model specifications. Representation A: mean + SD + Q10/Q50/Q90 per feature (460-dim). Representation B: softmax-weighted mean with tuned temperature τ (92-dim). Representation C: mean, max, Q90 of nucleus scores (3-dim). All tuning parameters selected by internal 5-fold CV within each LOOCV training fold.

Model	Representation	Classifier	Dim.
M0	None (majority class)	Always predict NR	—
M1	A	Lasso logistic regression (λ tuned)	460
M2	A	Random Forest (500 trees)	460
M3	A	PCA + LDA (M PCs, CV-tuned)	$\rightarrow M$
M4	A	Neural network (H , decay tuned)	460
M5	B (τ tuned)	Lasso logistic regression	92
M6	B	Random Forest	92
M7	B	LDA	92
M8	B	Neural network	92
M9	C	Lasso logistic regression	3
M10	C	Random Forest	3
M11	C	LDA	3
M12	C	Neural network	3
META	All (M1–M12)	Ridge LR on 84×12 prob. matrix	12

2.4 Stacking Meta-Ensemble (Layer 3)

Out-of-fold predicted probabilities from M1–M12 form an 84×12 meta-feature matrix. A ridge logistic regression meta-learner is fitted on this matrix and evaluated by a second LOOCV loop [8]. Ridge preserves all 12 base-model contributions; using out-of-fold predictions prevents the information leakage that would arise from training-set

predictions.

2.5 Evaluation Protocol

2.5.1 Patient-Level Leave-One-Out Cross-Validation

All 13 models are evaluated by patient-level LOOCV (84 folds). In each fold, the following steps are performed on the 83 training patients only, with the held-out patient accessed only at the prediction step:

- (1) Feature means and SDs estimated from training nuclei; applied to all patients for standardization.
- (2) Nucleus-level Random Forest scorer retrained from scratch on training-fold nuclei.
- (3) All tuning parameters (λ , H , M , τ) selected by internal 5-fold CV on 83 training patients.
- (4) Patient representations constructed using training-fold parameters.
- (5) Four classifiers fitted on 83 training patient vectors.
- (6) Held-out patient predicted; result stored.

This ensures no information about the held-out patient influences any aspect of the pipeline. Using patient-level labels as nucleus-level pseudo-labels introduces dependence *within* patients but not *across* patients, which is correctly handled by patient-level holdout.

2.5.2 Performance Metrics and Uncertainty Quantification

The primary metric is patient-level misclassification error:

$$\text{MCE} = \frac{1}{84} \sum_{i=1}^{84} \mathbf{1}(\hat{y}_i \neq y_i). \quad (6)$$

A 95% exact binomial (Clopper–Pearson) confidence interval is computed via `binom.test()` in R. Secondary metrics are sensitivity and specificity, reported separately to characterize the clinical tradeoff between missed recurrences and unnecessary interventions.

2.5.3 Statistical Comparison and Leakage Demonstration

McNemar’s test is applied to each pair of models with Bonferroni correction. To quantify leakage bias, M2 is additionally evaluated by 10-fold CV with nuclei randomly assigned to folds, and the difference in MCE from the valid patient-level estimate is reported.

Software. R version 4.3+; packages: `glmnet` [4], `randomForest` [6], `MASS`, `nnet` [7],

stats::prcomp.

3. Results

3.1 Overview

Table 2 summarizes LOOCV performance for all 13 models. The majority-class baseline M0 achieves MCE = 0.464 (95% CI: 0.355–0.576) with perfect specificity and zero sensitivity by construction.

Table 2. LOOCV performance for all 13 models. MCE = misclassification error; CI = 95% Clopper–Pearson exact binomial interval; Sens = sensitivity (TPR for R patients); Spec = specificity (TNR for NR patients). **Bold** = best value among M1–M12 and META. [†]Significantly worse than M0 (McNemar’s test, Bonferroni-adjusted $p < 0.05$).

Model	Description	MCE	95% CI	Sens	Spec
M0	Majority class (NR)	0.464	(0.355, 0.576)	0.000	1.000
M1	Lasso, Rep A	0.571	(0.459, 0.679)	0.231	0.600
M2	Random Forest, Rep A	0.607	(0.495, 0.712)	0.769	0.067
M3	PCA+LDA, Rep A	0.571	(0.459, 0.679)	0.410	0.444
M4	Neural Net, Rep A	0.560	(0.447, 0.668)	0.462	0.422
M5	Lasso, Rep B	0.429	(0.321, 0.541)	0.513	0.622
M6	Random Forest, Rep B	0.452	(0.343, 0.565)	0.538	0.556
M7	LDA, Rep B	0.500	(0.389, 0.611)	0.436	0.556
M8	Neural Net, Rep B	0.583	(0.471, 0.690)	0.282	0.533
M9 [†]	Lasso, Rep C	0.714	(0.605, 0.808)	0.231	0.333
M10 [†]	Random Forest, Rep C	0.714	(0.605, 0.808)	0.231	0.333
M11 [†]	LDA, Rep C	0.714	(0.605, 0.808)	0.231	0.333
M12 [†]	Neural Net, Rep C	0.714	(0.605, 0.808)	0.231	0.333
META	Stacking ensemble	0.321	(0.224, 0.432)	0.641	0.711

3.2 Primary Outcome: Misclassification Error

The stacking meta-ensemble achieved the lowest MCE at **0.321** (95% CI: 0.224–0.432), with sensitivity 0.641 and specificity 0.711, the only model to meaningfully outperform M0 on both metrics simultaneously (Figure 1). Among individual models, M5 (lasso on Representation B) achieved the best performance with MCE = 0.429 (95% CI: 0.321–0.541), sensitivity 0.513, and specificity 0.622.

Figure 1. LOOCV Misclassification Error with 95% Clopper-Pearson CI

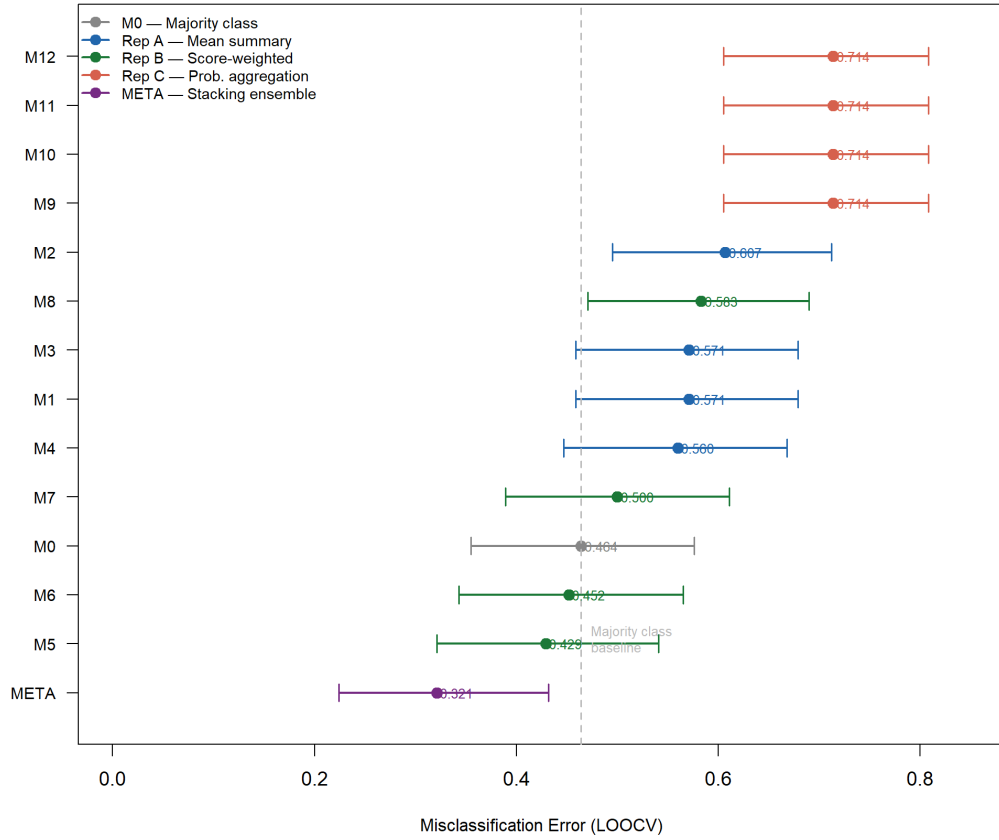


Figure 1. LOOCV misclassification error with 95% Clopper–Pearson confidence intervals for all 13 models, ordered from lowest (best) to highest MCE. **Blue:** Representation A (mean + quantile summaries, 460-dim); **green:** Representation B (score-weighted, 92-dim); **red:** Representation C (probability aggregation, 3-dim); **purple:** meta-ensemble. Dashed line: majority-class baseline (MCE = 0.464).

3.3 Effect of Patient Representation

A consistent pattern emerged across all four classifiers: Representation B outperformed Representation A, with the best Rep B model (M5, MCE = 0.429) improving over the best Rep A model (M4, MCE = 0.560) by 13.1 percentage points. This pattern

confirms that the aggregation strategy contributes more to predictive performance than classifier choice within a representation (see also Supplementary Figure 6). Representation C performed substantially worse, with all four classifiers achieving $\text{MCE} = 0.714$, significantly worse than the majority-class baseline (McNemar’s adjusted $p = 0.018$).

3.4 Effect of Classifier Choice

Within Representation B, lasso (M5, $\text{MCE} = 0.429$) and Random Forest (M6, $\text{MCE} = 0.452$) outperformed LDA (M7, $\text{MCE} = 0.500$) and the neural network (M8, $\text{MCE} = 0.583$). The neural network’s poorer performance reflects the small patient sample ($n = 84$). M2 (RF, Rep A) showed an anomalous pattern: high sensitivity (0.769) paired with near-zero specificity (0.067), indicating near-indiscriminate prediction of recurrence.

3.5 Sensitivity, Specificity, and Confusion Matrices

Figure 2 shows sensitivity and specificity for all models. In bladder cancer surveillance, sensitivity carries greater clinical weight than specificity, since a missed recurrence is clinically more harmful than an unnecessary follow-up. The meta-ensemble achieves the best balance (sensitivity 0.641, specificity 0.711). Figure 3 illustrates the progression: M0 identifies 0 of 39 recurrent patients; M5 identifies 20 of 39; META identifies 25 of 39, reducing total errors from 39 (M0) to 36 (M5) to 27 (META) out of 84 patients.

Figure 2. Sensitivity and Specificity by Model (LOOCV)

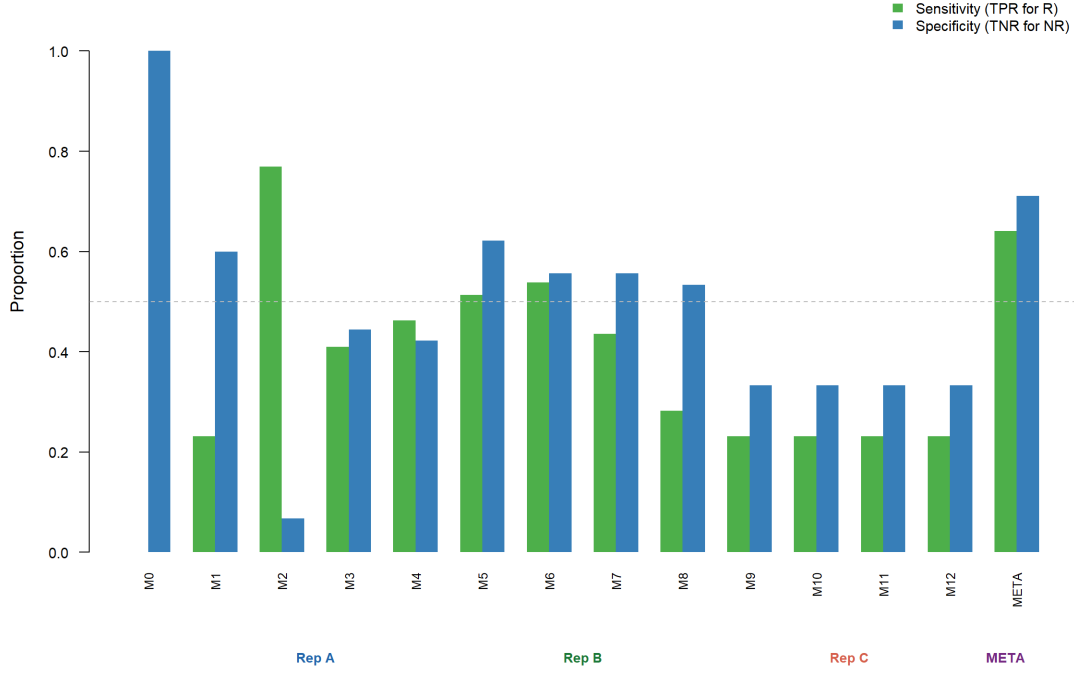


Figure 2. Sensitivity and specificity by model. Green: sensitivity (TPR for R patients). Blue: specificity (TNR for NR patients). Dashed line at 0.5. The meta-ensemble achieves the best clinical balance (sensitivity 0.641, specificity 0.711). M2 exhibits high sensitivity (0.769) but near-zero specificity (0.067), reflecting near-indiscriminate recurrence prediction. Representation C models (M9–M12) are uniformly poor on both metrics.

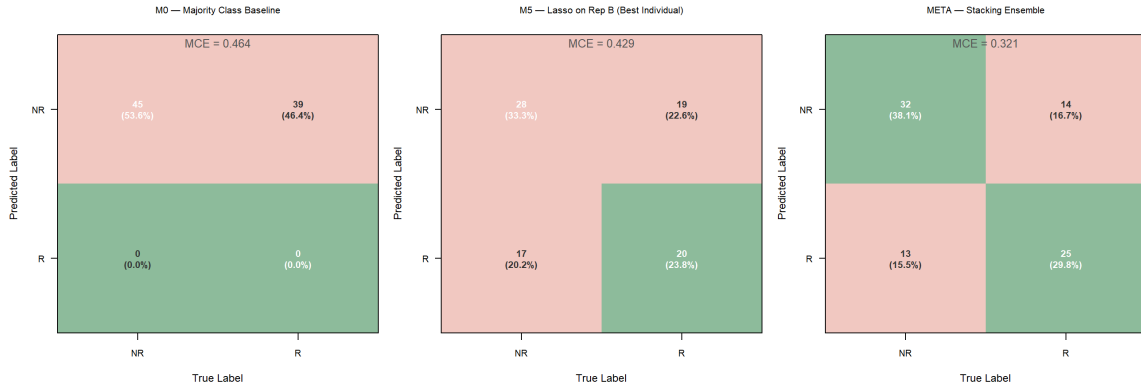


Figure 3. Confusion matrices for three key models. Green cells: correct classifications; pink cells: errors. *Left:* M0 — 0 of 39 recurrent patients identified; MCE = 0.464. *Centre:* M5 — 20 of 39 identified; MCE = 0.429. *Right:* META — 25 of 39 identified; MCE = 0.321. Total errors: 39 (M0), 36 (M5), 27 (META).

3.6 Additional Diagnostics

No individual model significantly outperformed M0 after Bonferroni correction (all adjusted $p > 0.05$); the meta-ensemble approached significance (unadjusted $p = 0.074$), consistent with limited power at $n = 84$. Nucleus-level 10-fold CV yielded an apparent MCE of 0.268 for M2 vs. the valid LOOCV estimate of 0.607, a bias of **0.339** (56% overoptimism; Supplementary Figure 5). Permutation importance (Supplementary Figure 4) identified variability summaries `f54_sd`, `f53_sd`, and `f53_q90` as the top features, confirming that nuclear heterogeneity is more prognostic than mean morphology; near-uniform softmax weights ($R = 0.0112$, $NR = 0.0118$) indicate this signal is diffuse across the nuclear population rather than concentrated in outlier cells.

4. Discussion and Conclusions

4.1 Principal Findings

This study provides three principal findings.

Aggregation strategy matters more than classifier choice. The score-guided softmax-weighted representation (Representation B) improved over mean-summary representations (Representation A) by 10–13 percentage points in MCE across all four classifiers, while classifier differences within a representation were smaller. This finding directly answers the study’s primary question: how nuclei are aggregated contributes more to patient-level predictive performance than which classifier is applied. It also validates the course’s suggestion that variability and percentile measures outperform simple means for this type of data.

Ensemble combination provides additional gain. The stacking meta-ensemble (MCE = 0.321) improved over the best individual model (M5, MCE = 0.429) by 10.8 percentage points, correctly identifying 25 of 39 recurrent patients. This demonstrates that different representations and classifiers capture complementary aspects of the data and that their combination via a simple ridge logistic meta-learner adds meaningful value.

Nucleus-level cross-validation is severely biased. The 33.9 percentage-point inflation from nucleus-level CV (56% overoptimism) represents a fundamental methodological flaw present in much of the karyometry literature. Patient-level LOOCV is the

only valid evaluation design for this hierarchical data structure, and its importance is reinforced empirically by this demonstration.

4.2 Interpretation of Feature Importance

The dominance of f54 and f53 variability measures (likely co-occurrence or run-length matrix statistics capturing chromatin texture) suggests that nuclear morphological heterogeneity is more prognostically informative than mean morphology. This is biologically plausible: cancer recurrence may be associated with cellular heterogeneity consistent with clonal evolution in bladder cancer, rather than uniform morphological shift. The diffuse nucleus-level saliency (near-uniform softmax weights) further suggests that recurrence is a population-level property of the tumor’s nuclear landscape rather than a feature of specific outlier cells.

4.3 Limitations

Sample size. With $n = 84$ patients, no individual model achieves statistical significance over the baseline after Bonferroni correction. All findings are descriptive and exploratory pending external validation on an independent cohort.

Probability calibration. Predicted probabilities were compressed near the marginal rate (≈ 0.46) due to strong regularization at $n = 84$. ROC-based analysis was not feasible; all comparisons rest on classification decisions at the 0.5 threshold.

Feature interpretation. The mapping of f1–f92 to named morphometric measurements was unavailable, limiting biological interpretation of top features.

MIL approximation. Using patient-level labels as nucleus-level pseudo-labels introduces noise for heterogeneous patients. End-to-end attention MIL [5] would be more principled but requires sample sizes an order of magnitude larger.

References

- [1] Korde, V.R., Bartels, H.G., Barton, J.K., & Ranger-Moore, J. (2009). Automatic segmentation of cell nuclei in bladder and skin tissue for karyometric analysis. *Analytical and Quantitative Cytology and Histology*, 31(2), 83–89.
- [2] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
- [3] Dietterich, T.G., Lathrop, R.H., & Lozano-Pérez, T. (1997). Solving the multiple instance problem with axis-parallel rectangles. *Artificial Intelligence*, 89(1–2), 31–71.
- [4] Friedman, J., Hastie, T., & Tibshirani, R. (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, 33(1), 1–22.
- [5] Ilse, M., Tomczak, J.M., & Welling, M. (2018). Attention-based deep multiple instance learning. *Proceedings of ICML 2018*. PMLR.
- [6] Liaw, A., & Wiener, M. (2002). Classification and regression by randomForest. *R News*, 2(3), 18–22.
- [7] Venables, W.N., & Ripley, B.D. (2002). *Modern Applied Statistics with S* (4th ed.). Springer.
- [8] Wolpert, D.H. (1992). Stacked generalization. *Neural Networks*, 5(2), 241–259.
- [9] Campanella, G., et al. (2019). Clinical-grade computational pathology using weakly supervised learning on whole slide images. *Nature Medicine*, 25(8), 1301–1309.
- [10] Nowell, P.C. (1976). The clonal evolution of tumor cell populations. *Science*, 194(4260), 23–28.

Supplementary Material

*From Nuclei to Patients: A Systematic Evaluation of Aggregation Strategies for
Bladder Cancer Recurrence Prediction*

S1. Supplementary Methods

S1.1 Neural Network Weight Budget

To satisfy `nnet`'s weight limit, hidden units are capped at $H_{\max} = \lfloor (3000 - 1)/(p + 2) \rfloor$ for input dimension p . For Representation A ($p = 460$), $H_{\max} = 6$; for Representation B ($p = 92$), $H_{\max} = 10$. Five random restarts per (H, λ) ; best solution retained by training loss.

S1.2 Handling Near-Zero-Variance Features in LDA

Features f13–f23 and f27–f44 are identically zero for the majority of nuclei. Before LDA, columns with within-class SD $< 10^{-8}$ are removed. If fewer than two features remain, the majority-class prediction is returned. All PCA+LDA inner-CV steps use `tryCatch` to handle remaining singularities.

S2. Supplementary Figures

Figure 4. Top 20 Karyometric Features by Permutation Importance
(Random Forest on Representation A, all 84 patients)

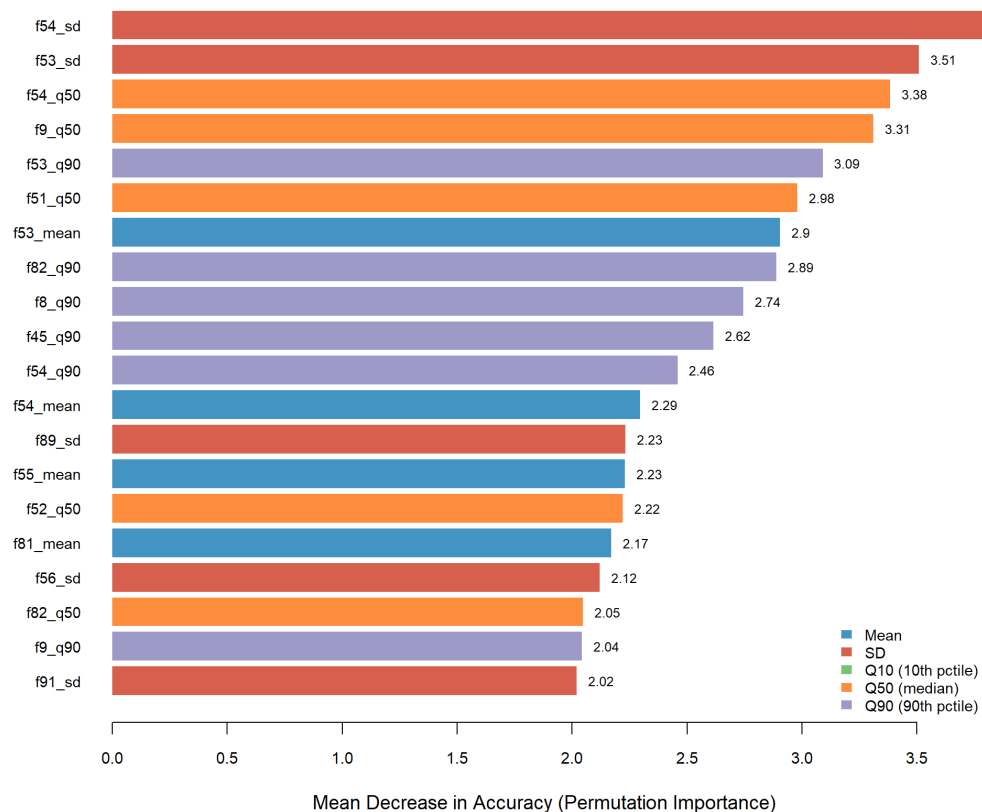


Figure 4. Figure S1. Top 20 karyometric features by permutation importance (Random Forest on Representation A, all 84 patients, 500 trees). Colour by summary statistic: **blue** = mean, **red** = SD, orange = Q50 (median), purple = Q90. Features f54 and f53 dominate; the prevalence of SD and Q90 summaries confirms that within-patient nuclear variability is more predictive than mean morphology.

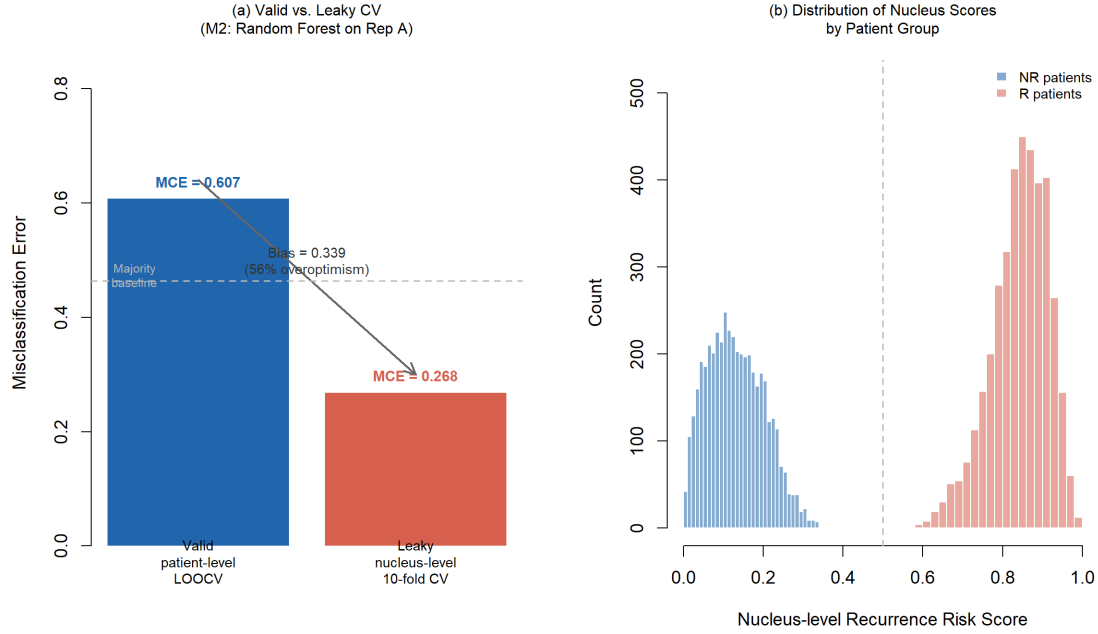


Figure 5. Figure S2. Leakage demonstration. *Panel (a):* M2 under valid patient-level LOOCV ($MCE = 0.607$, blue) vs. leaky nucleus-level 10-fold CV ($MCE = 0.268$, red). Leakage bias = 0.339 (56% overoptimism). Dashed line: majority-class baseline. *Panel (b):* Nucleus-level recurrence risk scores by patient group. NR nuclei score near 0 (blue); R nuclei score near 0.8–0.9 (red). The strong nucleus-level separation explains the leakage: the scorer memorizes patient patterns, making held-out nuclei from the same patient trivially classifiable.

Figure 7. LOOCV MCE: Representation $\times$ Classifier

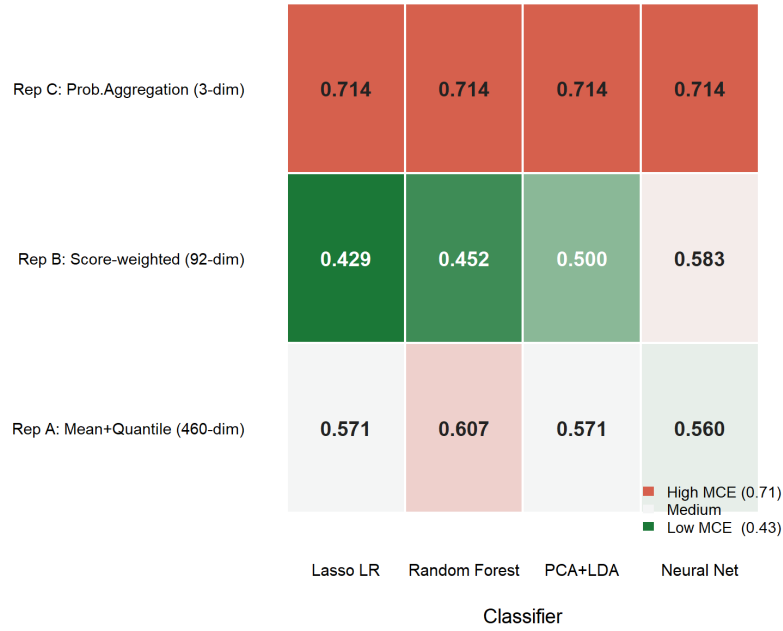


Figure 6. Figure S3. MCE heatmap: Representation $\times$ Classifier for M1–M12. Green = low MCE (better); red = high MCE. Representation C row (top): uniformly red (MCE = 0.714). Representation B row (middle): consistently greener than Representation A (bottom); best cell is M5, Lasso + Rep B (MCE = 0.429). Variation across rows (representations) exceeds variation across columns (classifiers), confirming aggregation strategy as the primary performance driver.